

Homework 2 - Extra Credit: Shor's Algorithm (10 points)

Group 3

A. Basis of Shor's algorithm and its relation to Pollard-Rho (5 points)

Shor's algorithm uses repeating patterns in modular arithmetic. For factoring a composite number N , it chooses a number a that is relatively prime to N and finds its order r . This means r is the smallest positive number for which $a^r \bmod N = 1$. If r is even and $a^{(r/2)}$ is not 1 or -1 modulo N , the ordinary calculations $\gcd(a^{(r/2)} - 1, N)$ and $\gcd(a^{(r/2)} + 1, N)$ give factors of N . If those conditions fail, the algorithm can try another a . Shor also gives a related method for a discrete logarithm problem such as $Y = g^x \bmod p$. It uses a repeating relation between g and Y to find the unknown x . (Shor, 1995)

Pollard-Rho also attacks discrete logarithms, but it follows a classical sequence of values until two values collide. That collision gives an equation that can be solved for x . In a group with $p - 1$ elements, its expected work is roughly proportional to the square root of $p - 1$. Shor uses a quantum procedure to find the repeating relation and needs a number of steps that grows polynomially with the number of input bits on a suitable quantum computer. Both methods use properties of modular arithmetic, but they find different kinds of patterns. (Pollard, 1978; Shor, 1995)

B. Why Shor's algorithm is quantum (5 points)

A quantum computer uses qubits, which can be prepared in a superposition of many input values. Shor's algorithm performs modular arithmetic on that superposition. It then applies a quantum Fourier transform. Interference makes some measurement results more likely than others, and those results carry information about the period needed for factoring or the relation needed for a discrete logarithm. The measured result does not simply list every possible answer. It gives useful clues that ordinary calculations can finish. (Shor, 1995)

For factoring, the classical part uses the measured information to recover r and then calculates the two greatest common divisors described in part A. For a discrete logarithm, the measurements give relations that can be used to recover x . The algorithm may repeat the quantum part if a measurement is not useful. Superposition, interference, and the quantum Fourier transform are what make this a quantum algorithm rather than a classical search. (Shor, 1995)

References

Shor, Peter W. "Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer." 1995. <https://arxiv.org/abs/quant-ph/9508027>

Pollard, J. M. "Monte Carlo Methods for Index Computation (mod p)." *Mathematics of Computation* 32 (1978): 918-924. <https://www.ams.org/mcom/1978-32-143/S0025-5718-1978-0491431-9/S0025-5718-1978-0491431-9.pdf>